RISC-V MARKET TRENDS And Predictions Till 2030

Big players are turning to RISC-V, with plans for integration already in motion. Its adaptability in tailoring ISA extensions suits specific solutions in AI silicon development and multi-core SoC designs. Gaining momentum, RISC-V is challenging norms and democratising cutting-edge tech

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The kind of flexibility that RISC-V provides designers has led to a revolution in the semiconductor industry. It is no longer a single party playing field. Multiple cores within a chip now originate from multiple parties (third party IPs).

With increasing computational complexity and decreasing chip size, Dennard's scaling has begun to reach its limits. Dennard's scaling is a guide for optimised shrinking of a semiconductor chip to enhance performance without increasing power usage.

So, companies have transitioned to multiple cores on a single chip. IP cores within a chip used to originate from a single party, but that trend is changing. With the emergence of the open standard ISA, RISC-V, many vendors are now developing optimised cores.

What is RISC-V?

RISC-V is an open standard instruction set architecture (ISA), a guideline on how software can instruct a processor. It is based on reduced instruction set computer (RISC) architecture. The RISC-V movement started at the University of California, Berkeley, in 2010.

What's so special about RISC-V anyway?—Its openness. RISC-V ISA allows designers to customise and extend this ISA to add specialised instructions and meet specific requirements of an SoC design. Now, who doesn't want that!

RISC-V played a significant role in the post-Covid era's recovery and growth of the dynamic semiconductor market. It has revolutionised system-on-chip (SoC) design strategies across multiple sectors, sparking extensive exploration of CPU architectures. This has significantly affected device revenues, unit shipments, design starts, and IP licensing revenues worldwide. Recently, the SHD group released its RISC-V Market Analysis Report, which predicted how RISC-V will fulfil the ever-growing chip demand.

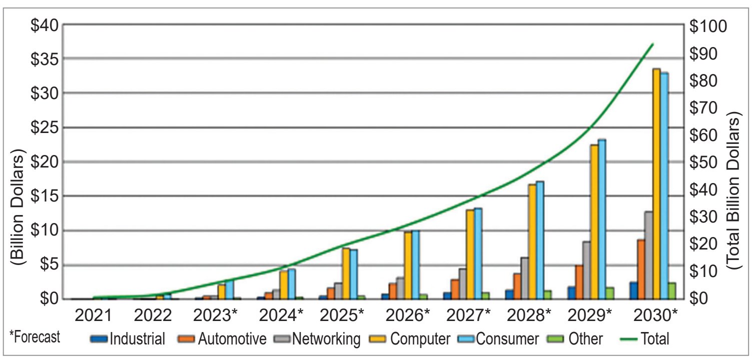

Fig. 1: Market revenues for all RISC-V SoCs by application, 2021-2030 (Source: The SHD Group, January 2024)

RISC-V-based SoC market analysis

Market revenue. The RISC-V SoC market reached $6.1 billion in 2023, a growth of 276.8% over 2022. It is projected to grow to $92.7 billion (47.4% CAGR) by 2030.

With increasing support from software and EDA tools, SoC designers continue to gain confidence in adopting RISC-V to design IP solutions. RISC-V-based CPU cores will be increasingly included alongside competing IPs, as more powerful RISC-V cores emerge.

Key segments showing growth in RISC-V inclusion are consumer electronics, reaching $2.8 billion in 2023, computers at $2.1 billion, automotive at $0.45 billion, networking at $0.43 billion, and industrial at $0.2 billion. The rapid adoption of RISC-V SoCs is driven by the integration of RISC-V CPU cores into